FUBAR LABS × 4-H

Fair Use Building and Research Labs

Introduction to Electronics and Breadboarding Circuits

2026

INTRO

What we're going to learn today:

01 What is an electronic circuit?

02 What kind of power is needed for these projects?

03 What are the fundamental principles of electronics?

04 What are the basic electronic components used in DC analog circuits?

05 How do these principles combine to make interesting things?

We're going to make some of those things, and you can take the kit, and the projects home with you.

Fundamentals

Reminder: the properties used in electronics were "found" or explained by many different people in the past. The contributions are part of the naming structure for the properties they documented.

PERSON	PROPERTY	MEASUREMENT
Alessandro Volta	Voltage	**Volts**
André-Marie Ampère	Amperes, current	**Amps**
James Watt	Watts, Power	**Watts**
Georg Ohm	Resistance	**Ohms**
Michael Faraday	Capacitance	**Farads**

INTRO

About Parts

Physics and chemistry in a tiny package

Symmetric



Asymmetric

Polarized



Identifying Parts

1

Review parts in kit

2

Physical Identification

3

Properties of parts

4

Data sheets document it all

INTRO · KIT OPTION

DIY Kit Parts List

DIY

REQUIRED · EVERY EXAMPLE

Breadboard	1
Wire kit	1
9V battery	1
9V snap power connector	1
Red LEDs	3
RGB LED	1
330Ω Resistors	3
1kΩ Resistors	1
Potentiometer 10kΩ	1
Button	2
Slide button, switch	1
Tilt ball switch	1
Reed switch	1
Magnet	1
Hall effect sensor	1
Transistor PN2222	1
Diode 1N4001	1
DC toy motor	1
1000µF capacitor 25V	2
100µF capacitor 25V	1

EXTRAS · OPTIONAL

Green LEDs	3
Yellow LEDs	3
10kΩ Resistors	3
10µF capacitor 25V	2
Potentiometer 10kΩ RGB mixing	2

Schematics use 330Ω for LEDs. Older breadboard pictures may show 220Ω: swap in 330Ω.

ELECTRICITY

What is Electricity?

What can it do?

What are the dangers?

What kinds are there?

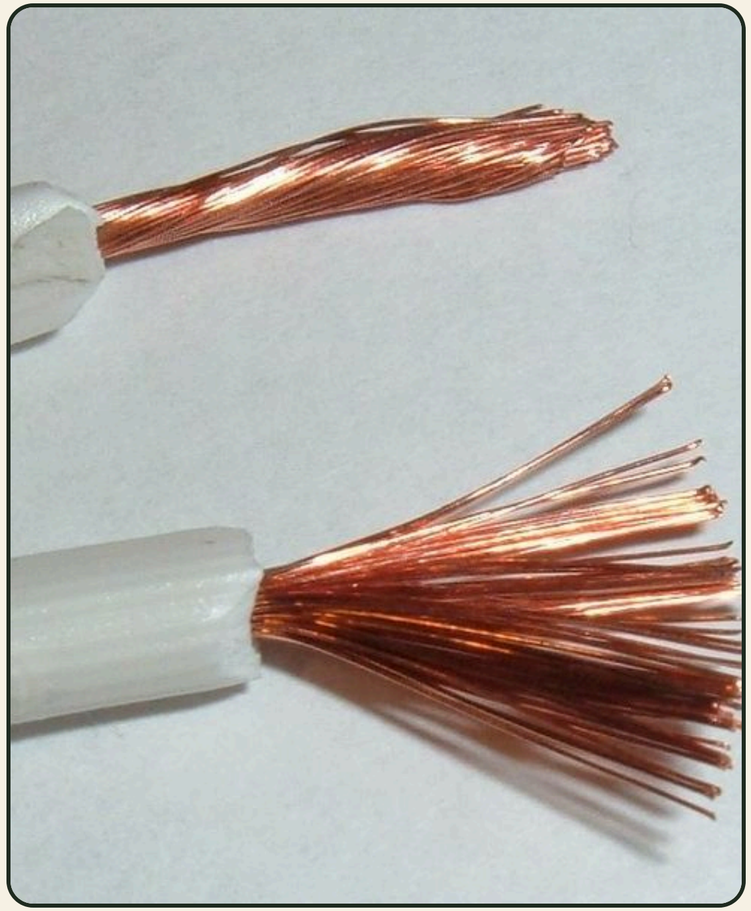
ELECTRICITY

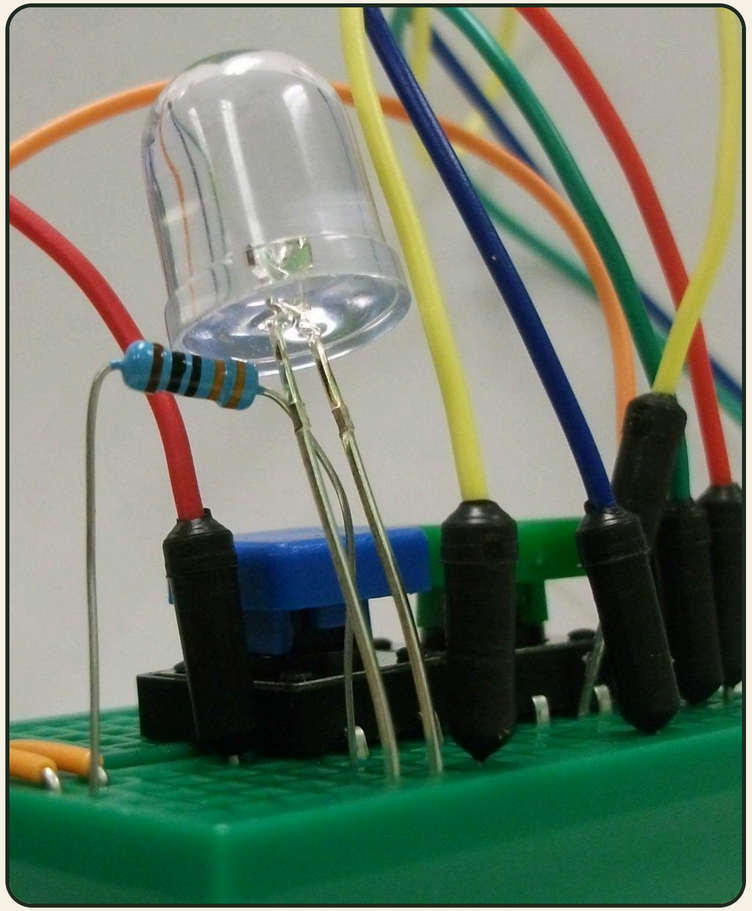
Insulators and Conductors

Wires are conductors of electricity

Different materials have different conductivity properties

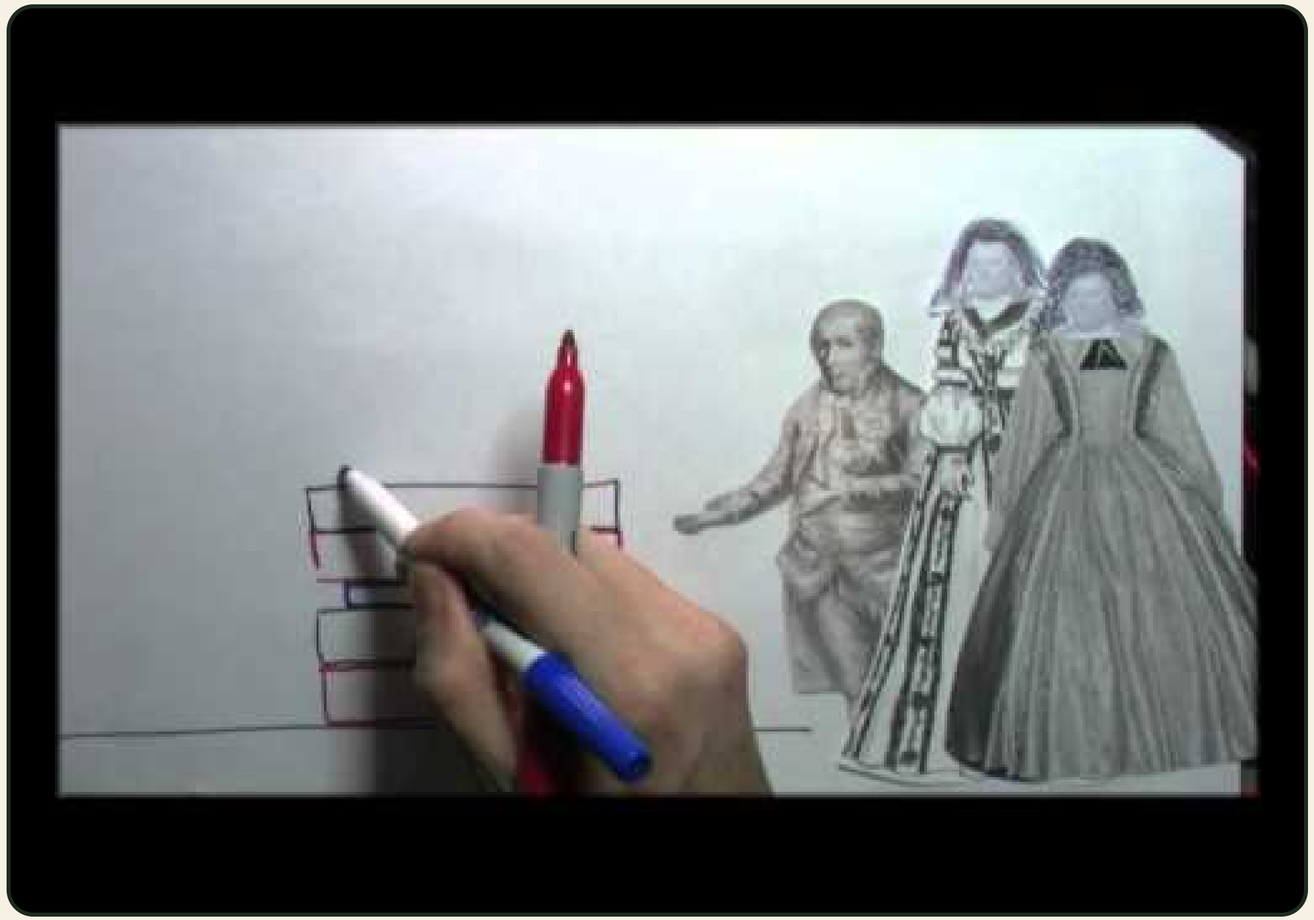
The plastic around wires is an **insulator**. They resist the flow of electricity





Batteries as a Power Source

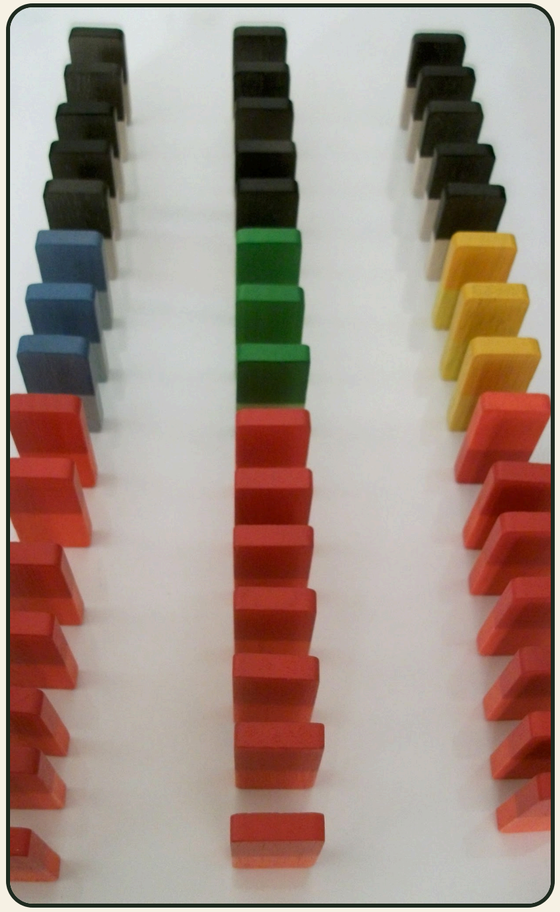
What kinds of batteries are there?

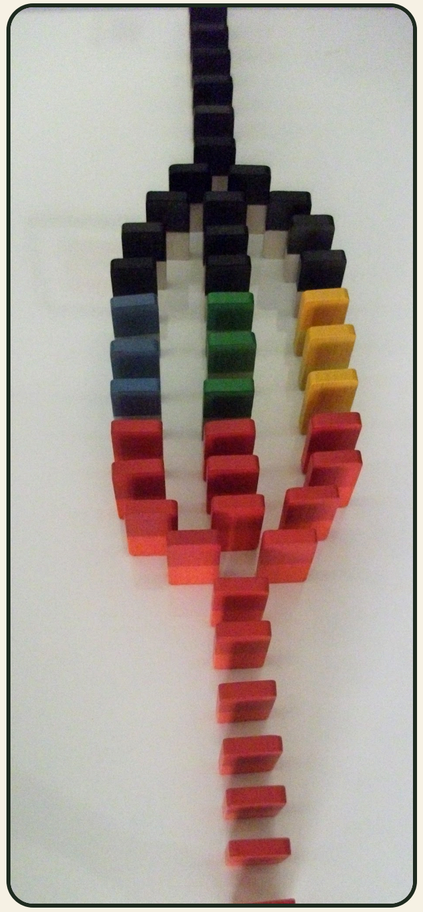


CIRCUITS

What is a circuit?

Combination of electronic parts, wires connected between power sources. It's like a physical program. It's also like setting up dominoes in sequence.



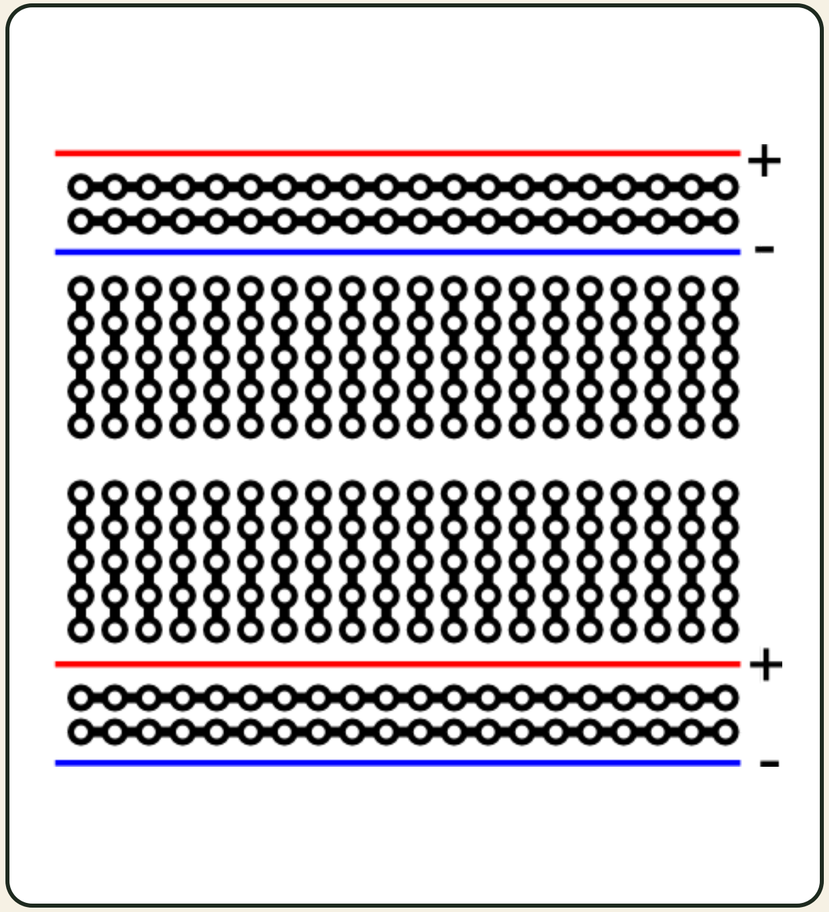


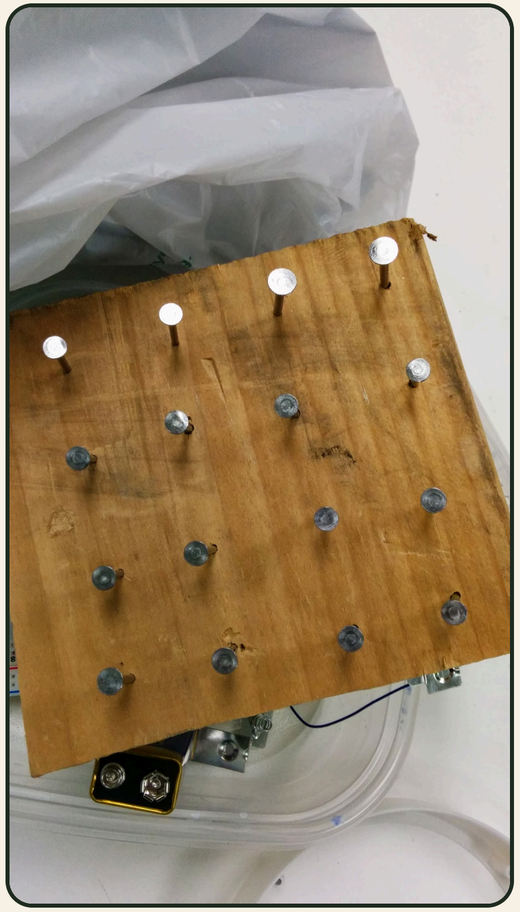
What is a Breadboard?

WHAT ARE THEY GOOD FOR?

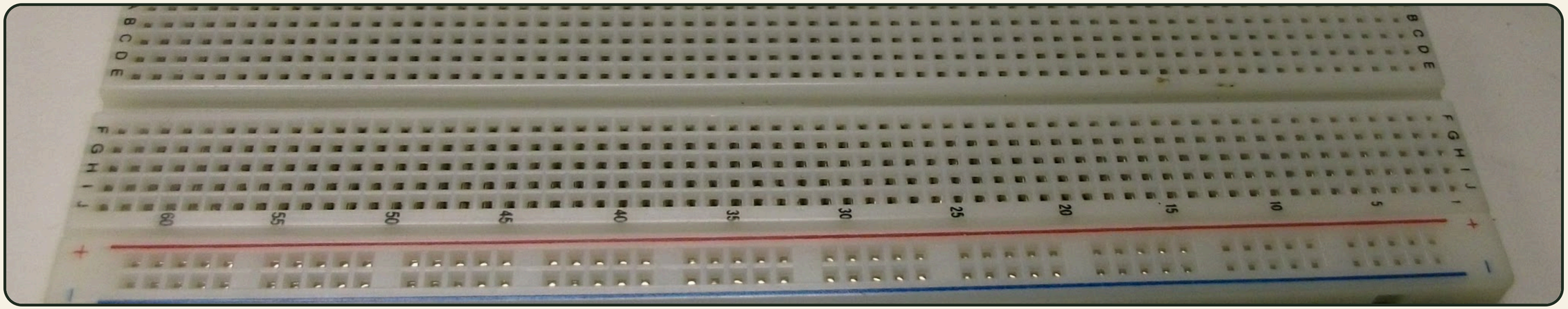
Creating, organizing, and prototyping a circuit.

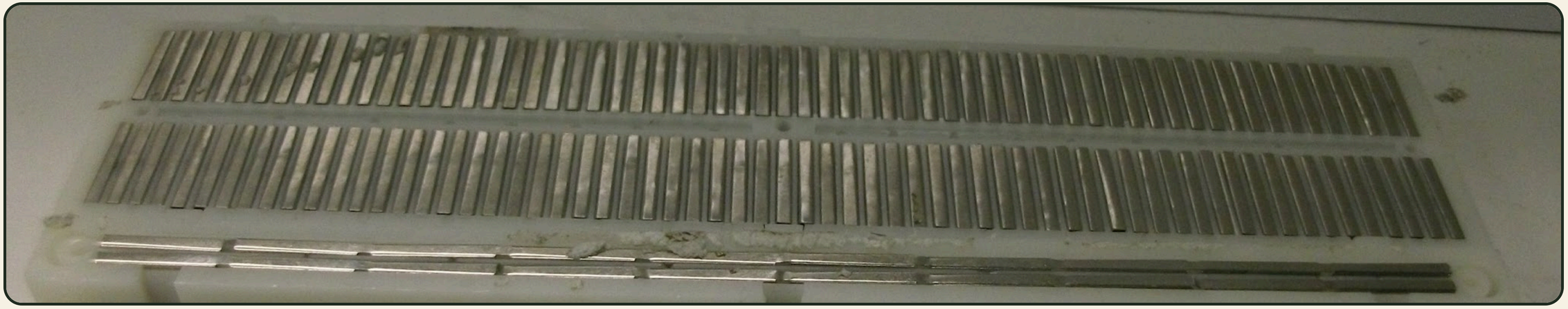
Literally started out as a bread board with nails.





Inside of a Breadboard



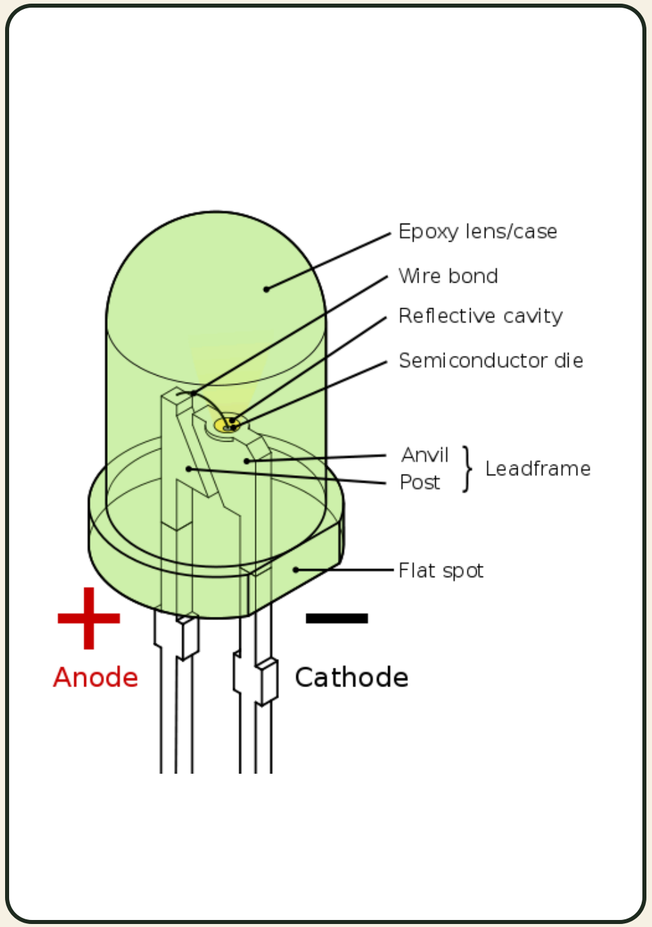


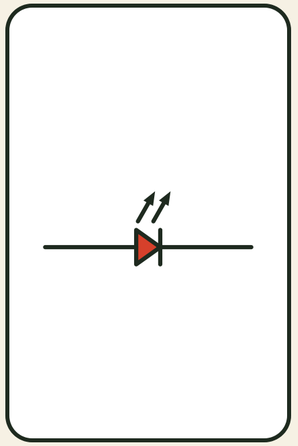
What are LEDs?

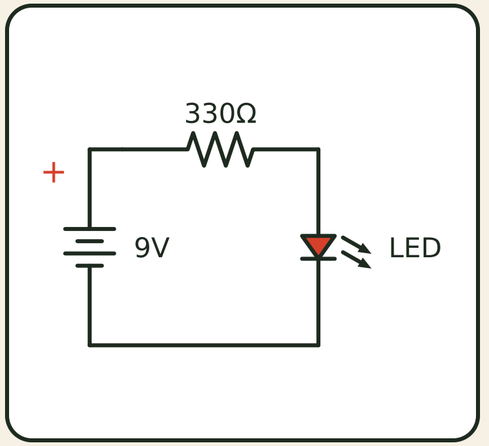
Light Emitting Diodes

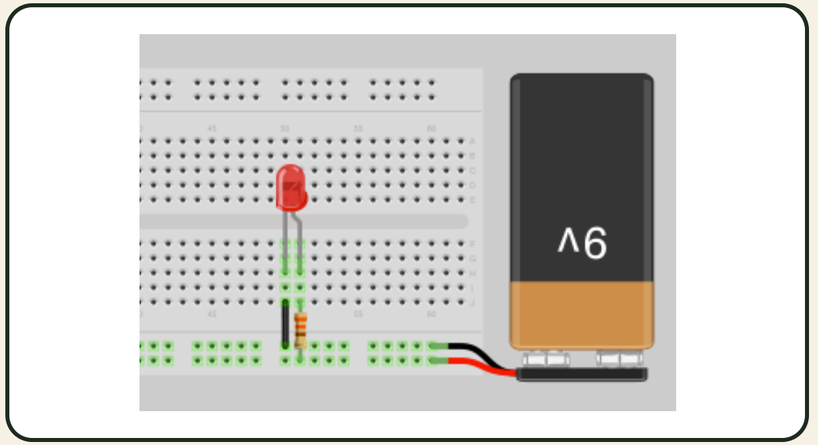
Diode Symbol + Arrows for light

Points to ground









PROJECT 1 · LIGHT AN LED

Hello World for a Circuit

PARTS

Battery

Resistor

LED

WHY

Power Source

An LED will light up when enough voltage is supplied but can also burn out if too much current is allowed to pass through. The resistor limits the current to prevent damage.

DO

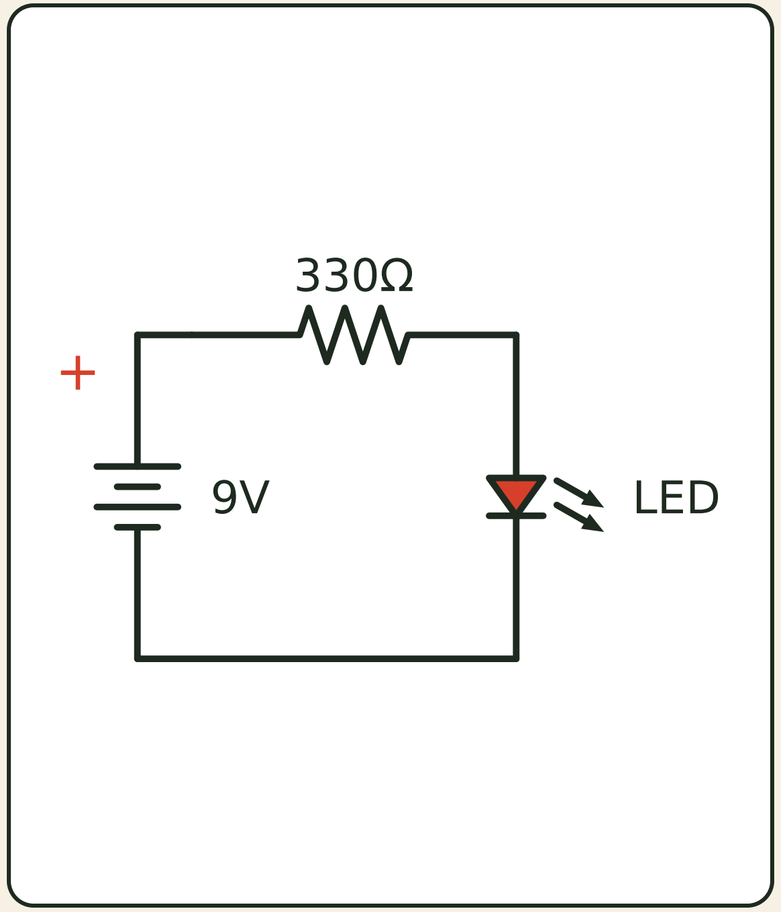
Connect Battery, Resistor, LED

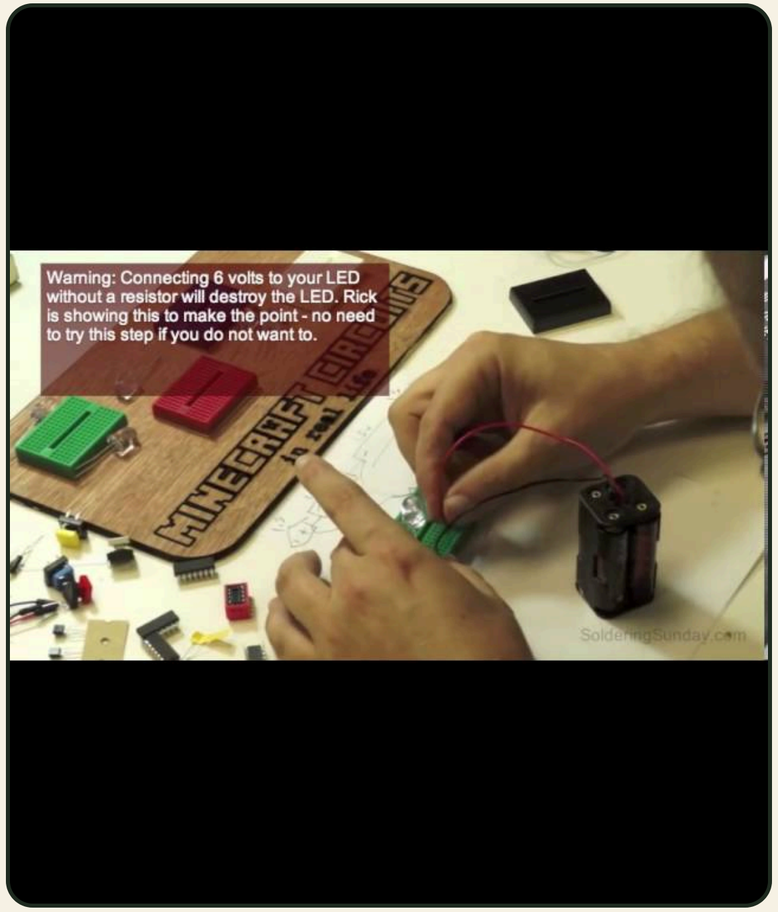
LED Circuit

1 LED Plus resistor

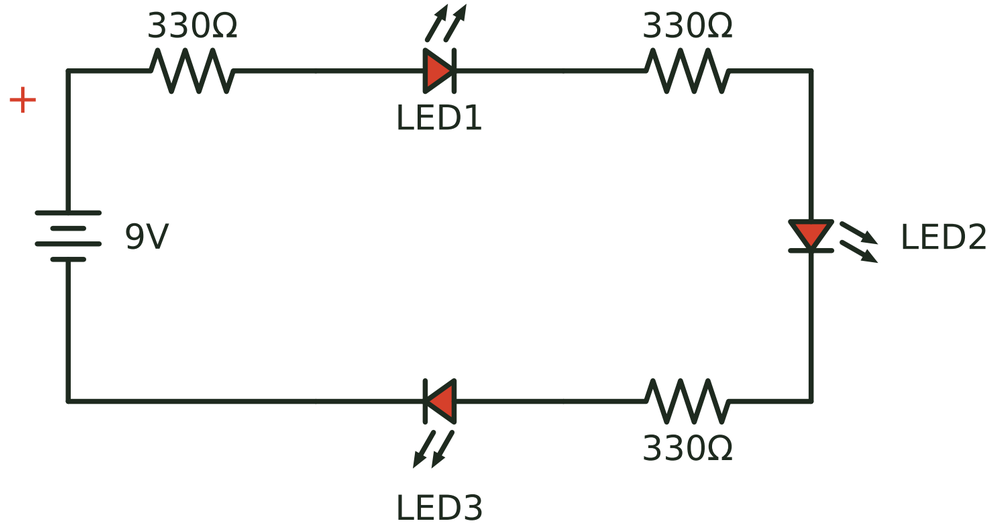
WHY HAVE A RESISTOR?

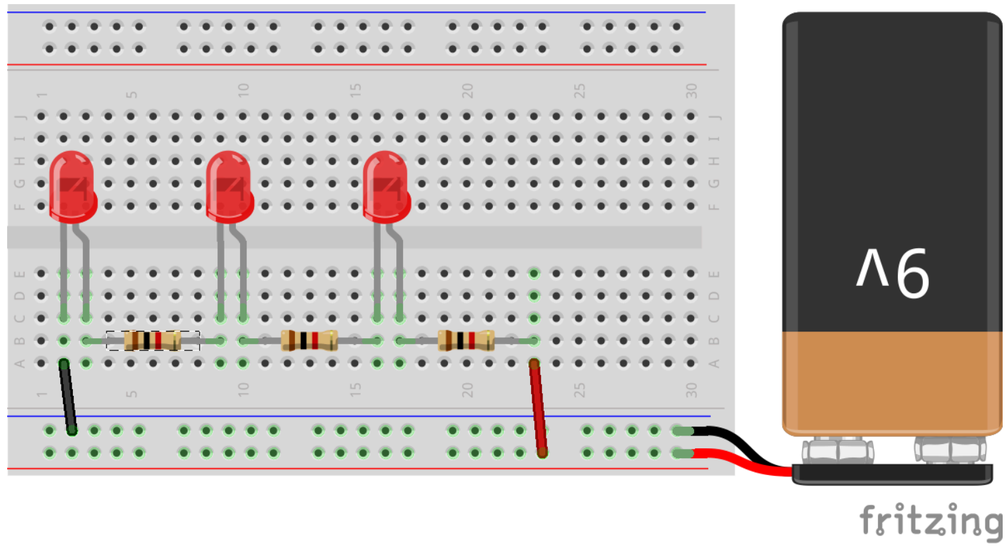
Current limiting.
Prevent LED burnout.



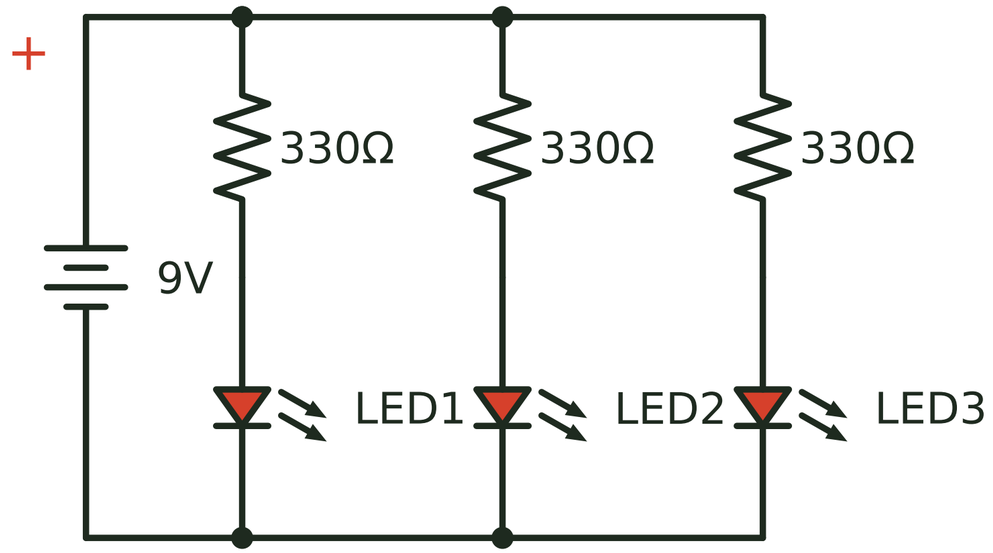


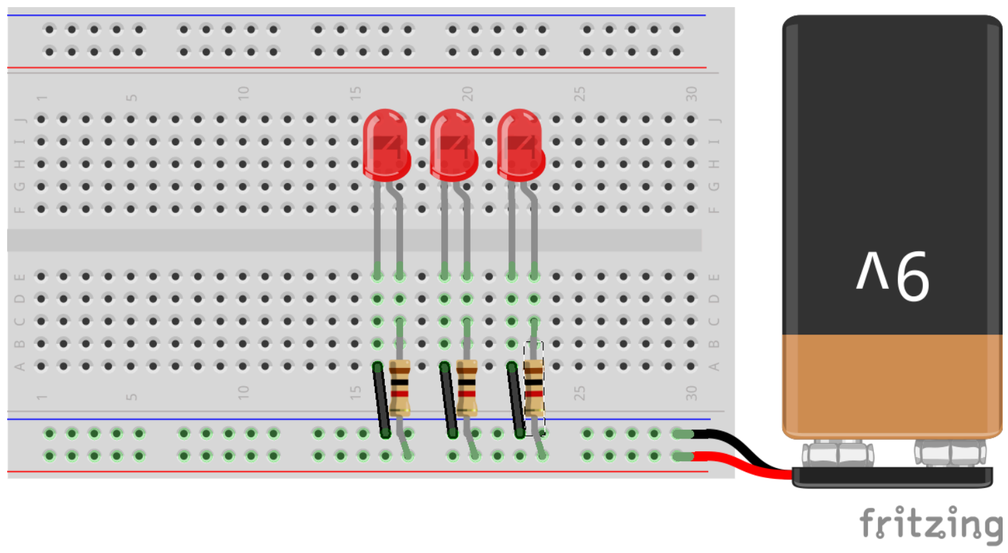
Series circuit example



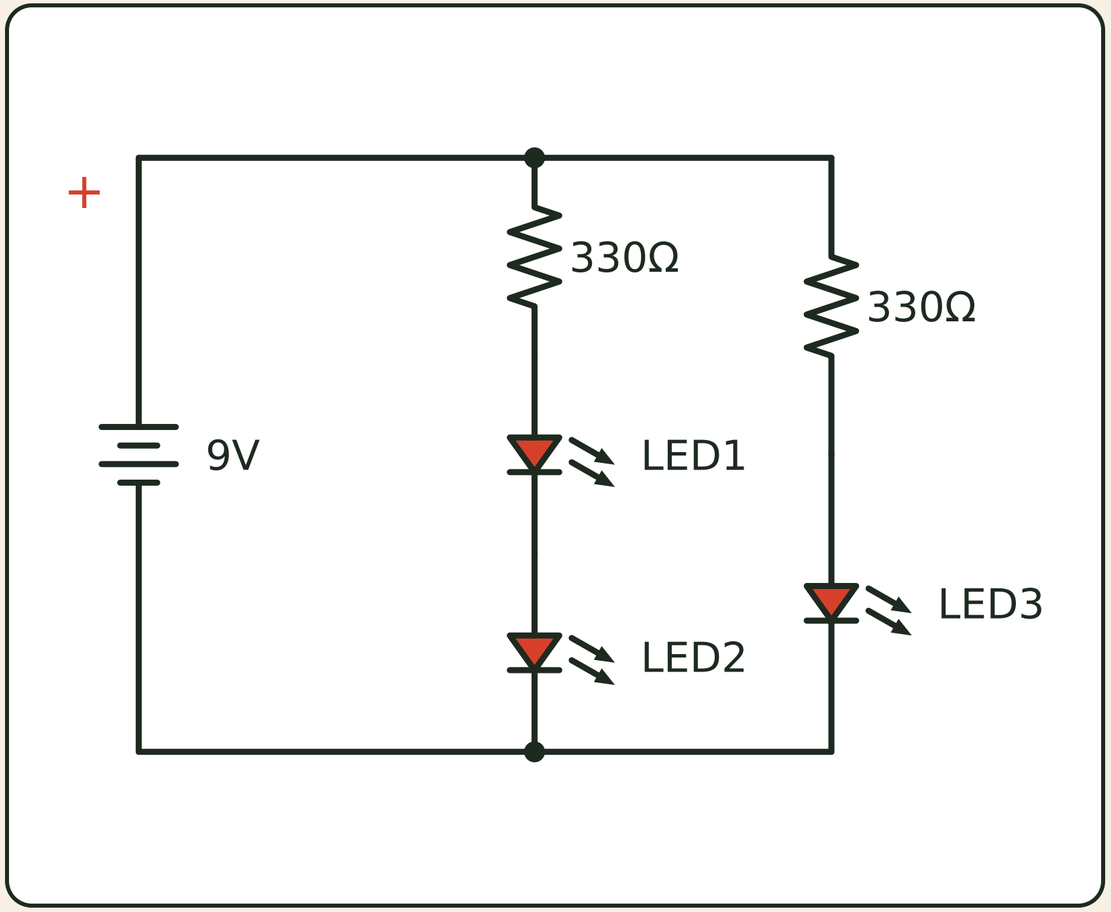


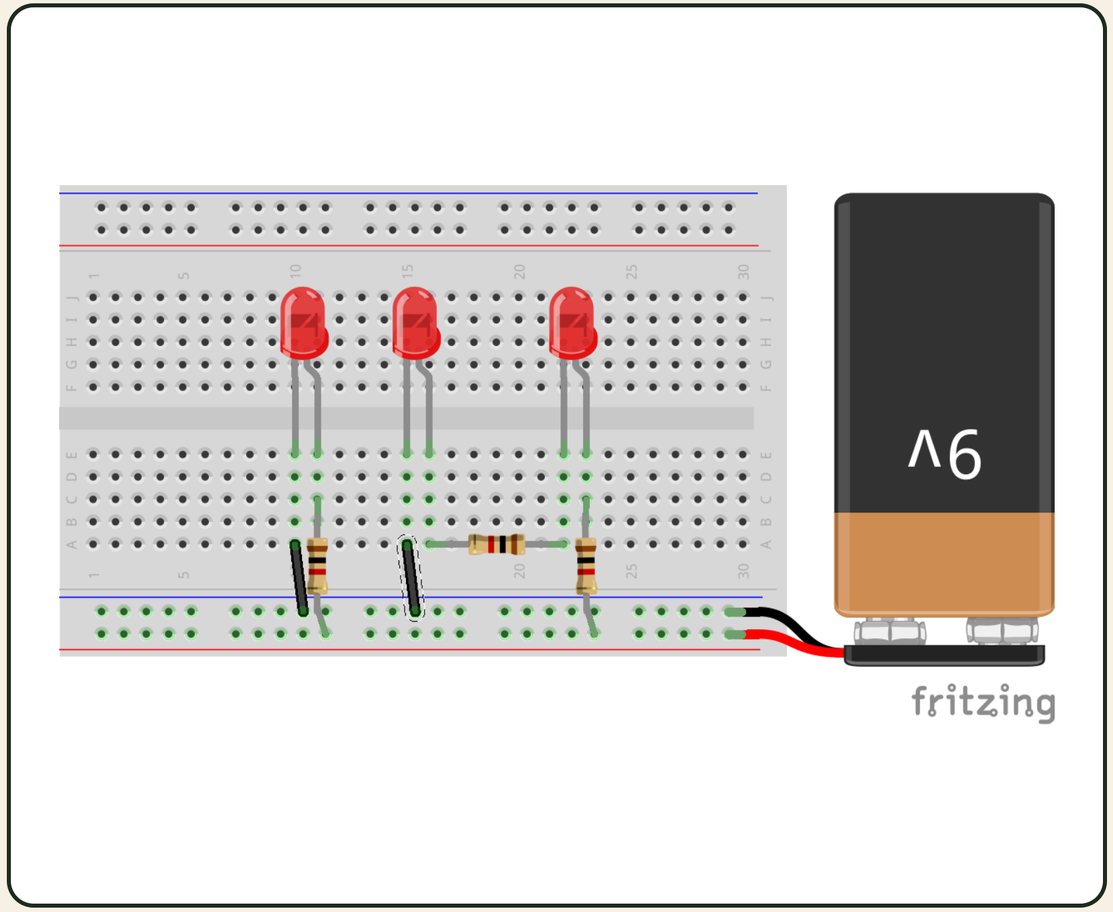
Parallel Circuit example





Parallel and Series Circuit Example





COMPONENTS

Fundamental Parts

- Resistor
- Capacitor
- Wire
- Diodes
- Light Emitting Diodes (LED)
- Battery
- Transistors
- Motors

They each have a physical property that interacts with electricity. When put in combination various actions occur.

What are Resistors?

Resistors provide a specific amount of resistance to a path in a circuit or wire. Ohm's law is used to calculate the properties related to resistance.

Resistors are color coded.

OHM'S LAW

$$I = V/R$$



I = Current measured in Amps

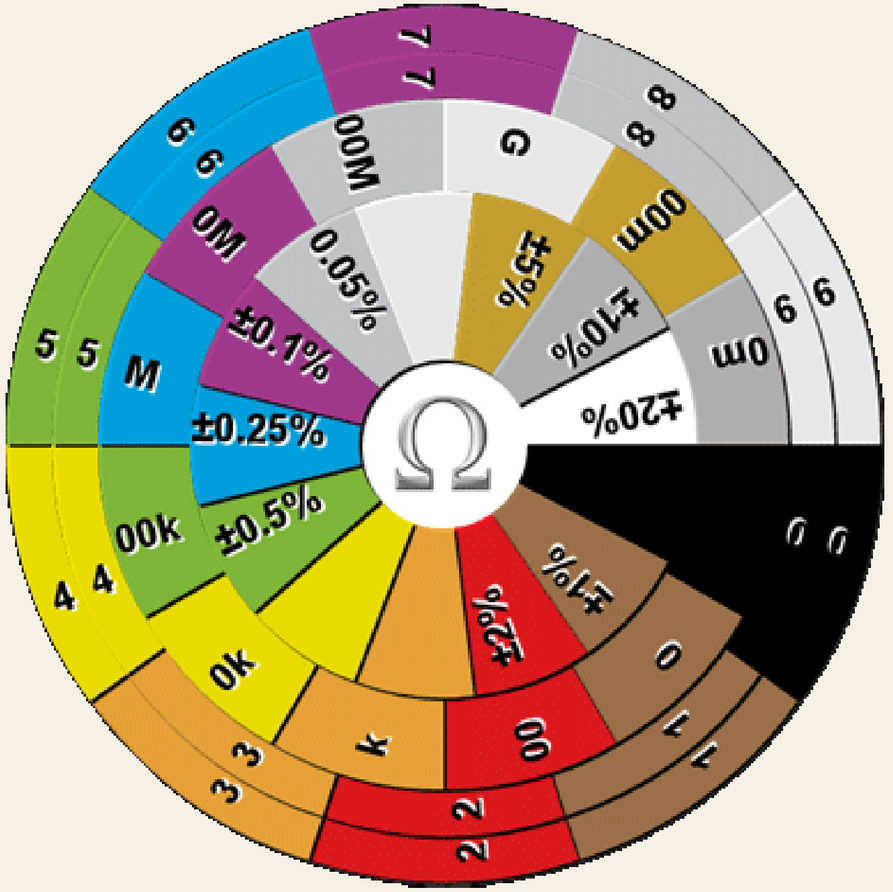
V = Voltage measured in Volts

R = Resistance measured in Ohms

RESISTORS

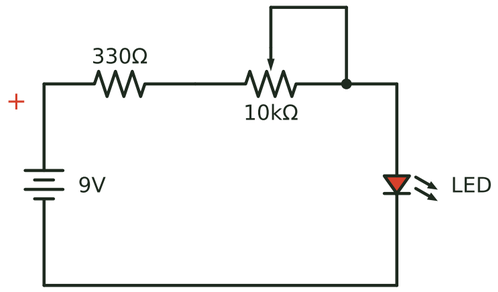
Resistor Color Code

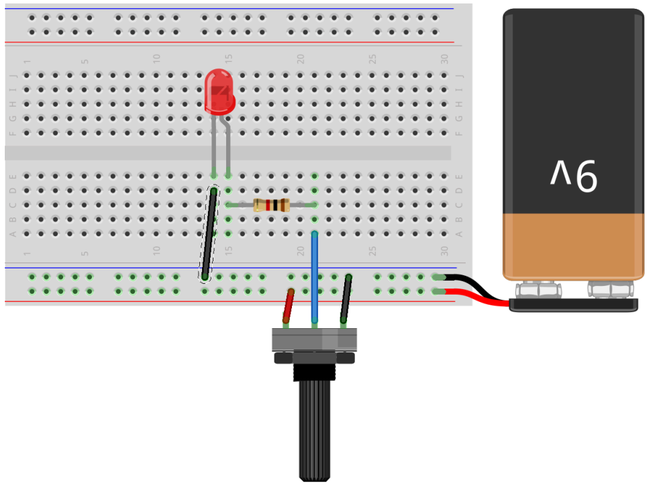
[Resistor Color Code Wheel Calculator]

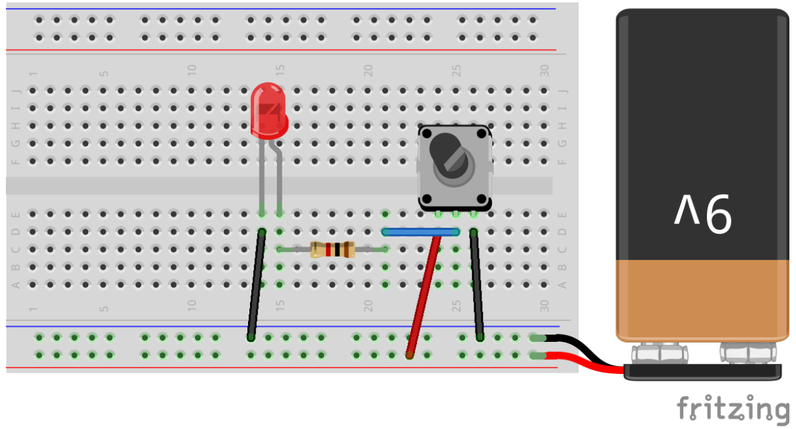


Variable Resistor: The Potentiometer

Voltage dividers

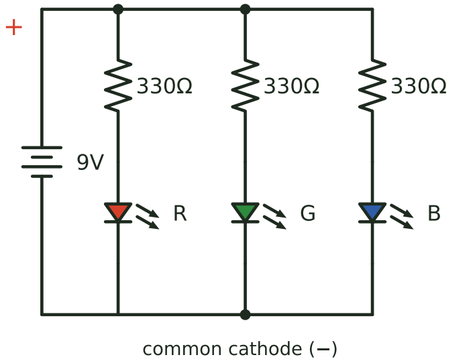




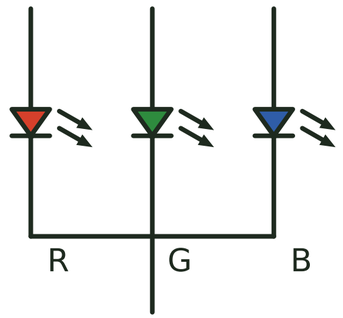


RGB LED

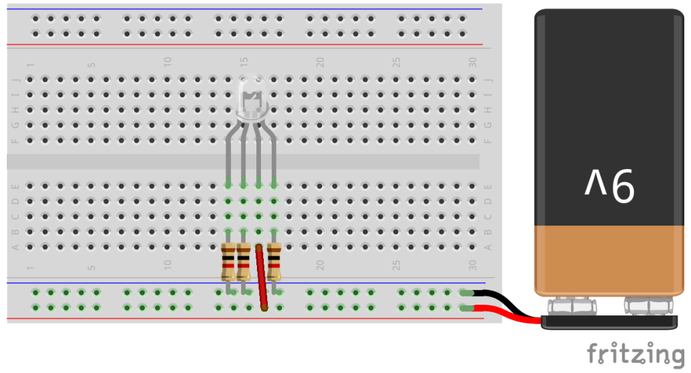
Try resistors out in various combinations to make different colors. Experiment by adding potentiometers to the leads.



common cathode (−)



common −



What are Capacitors?

A capacitor stores two separated charges.

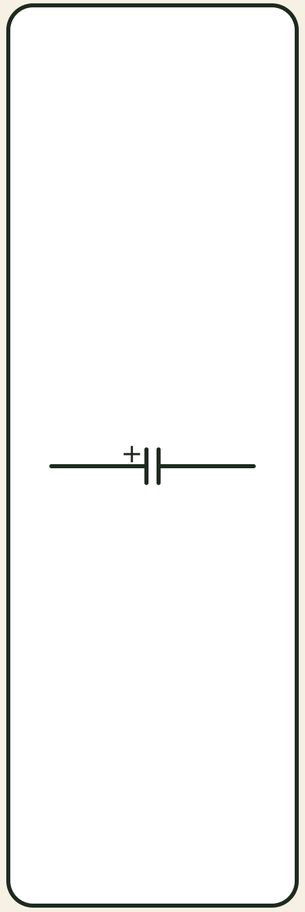
Known charge up time.

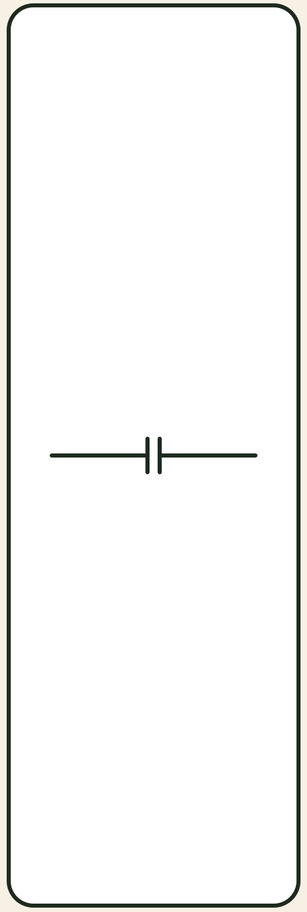
Known discharge time.

TWO MAJOR KINDS

Electrolytic, asymmetric, polarized

Ceramic, symmetric

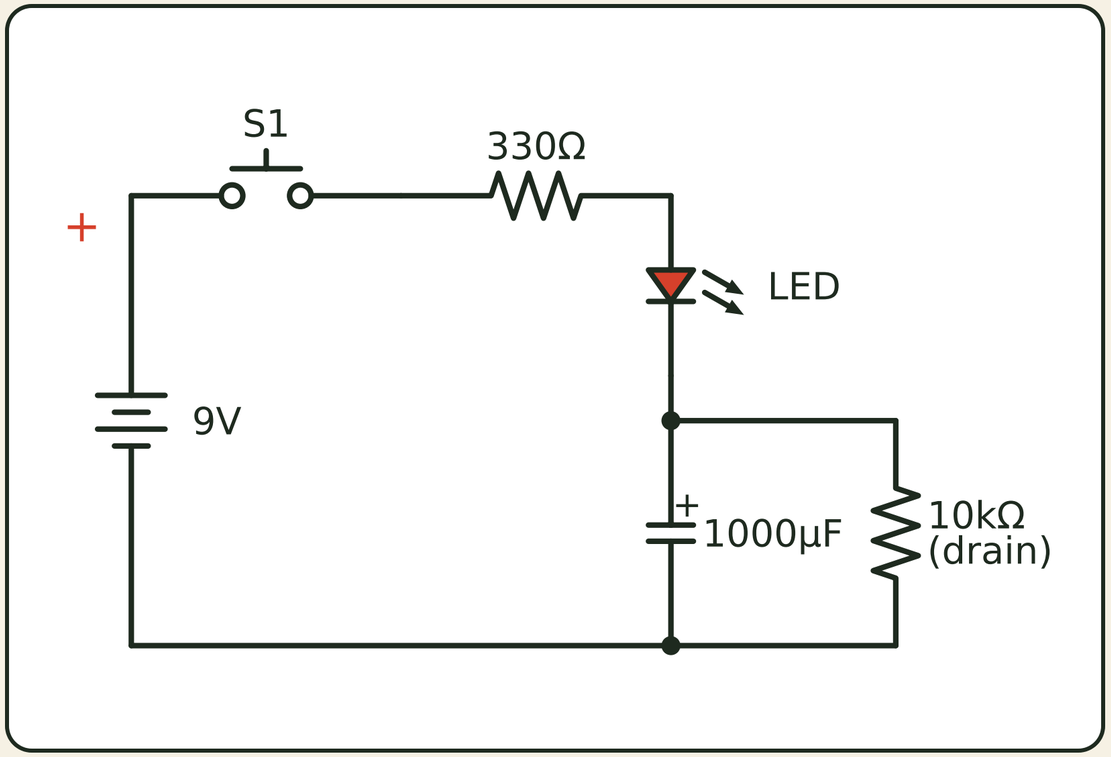


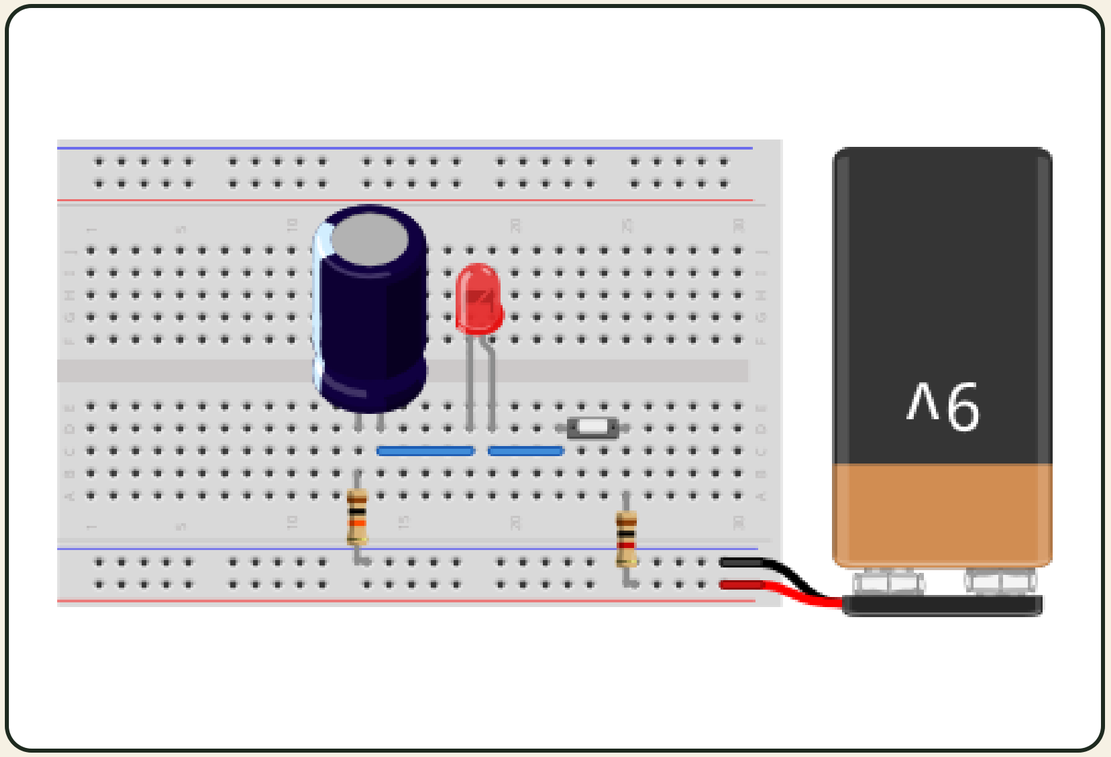




Capacitors in Series

Press S1: the LED flashes while C1 fills, then goes dark. The 10kΩ drains C1 so you can try again.



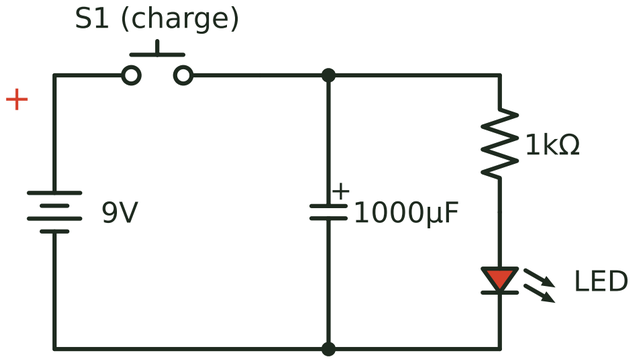


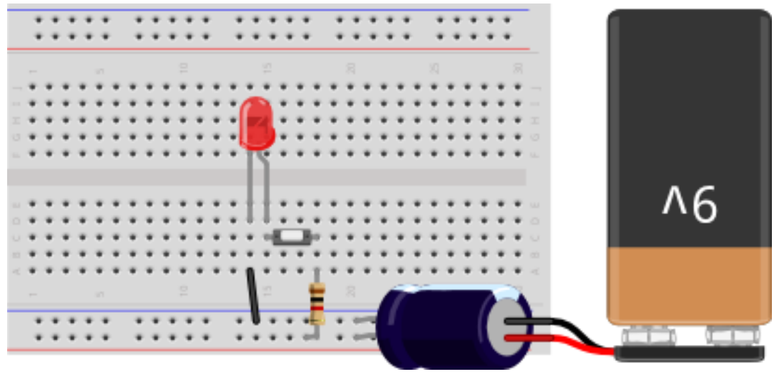
Capacitors in Parallel

They can provide energy, however briefly.

They can smooth out a signal.

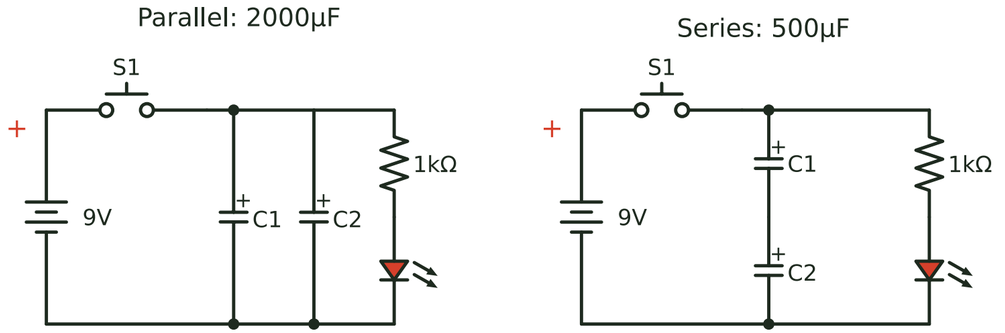
Hold S1 to charge C1. Let go and watch the LED fade as C1 empties.

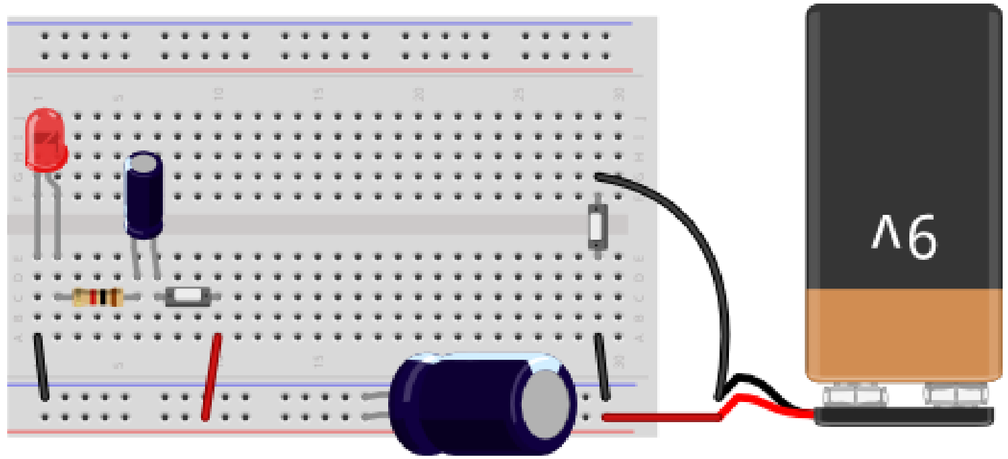




Capacitors in Series and Parallel

Two 1000µF caps: in parallel they add up (2000µF), in series they halve (500µF). Time each glow.





ACTIONS

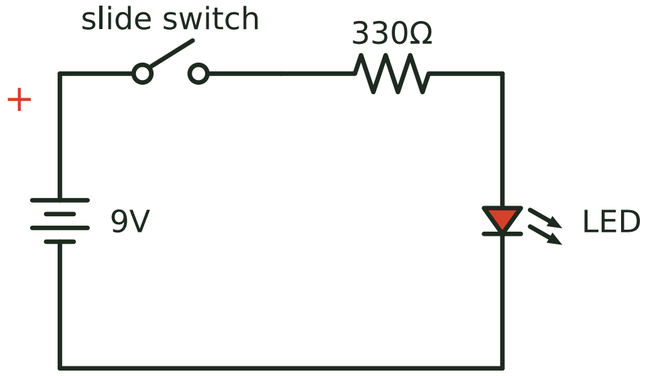
Turn things on and off

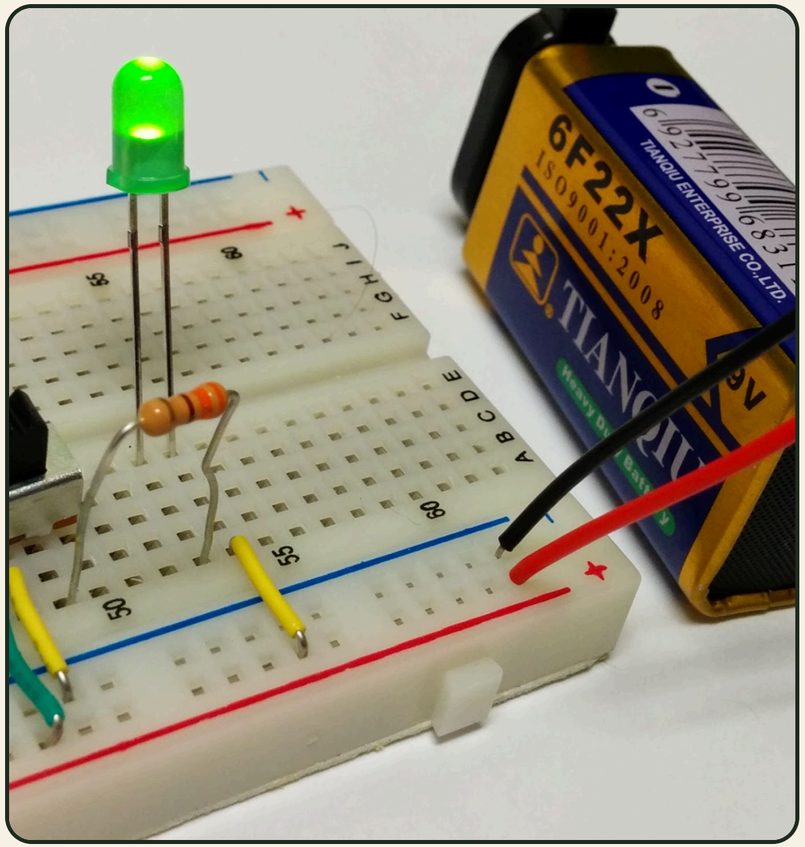
- Wire
- Reed Switch
- Hall Effect Sensor
- Relay
- Transistor
- Button
- Switch

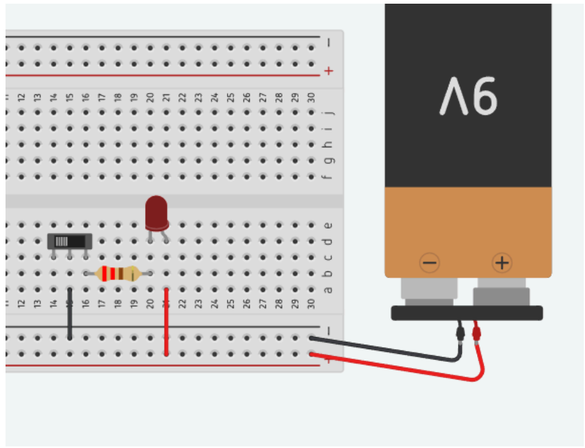
SWITCHES

The Slide Switch

Keeps its state until force applied.



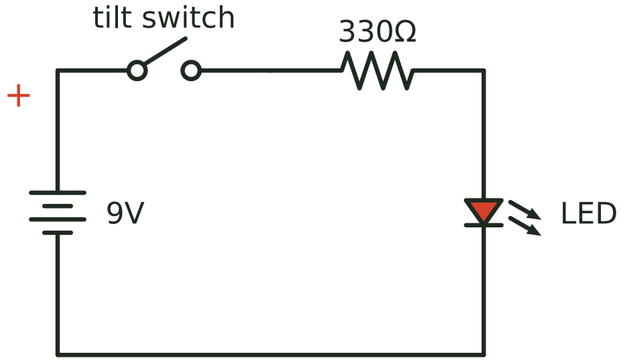


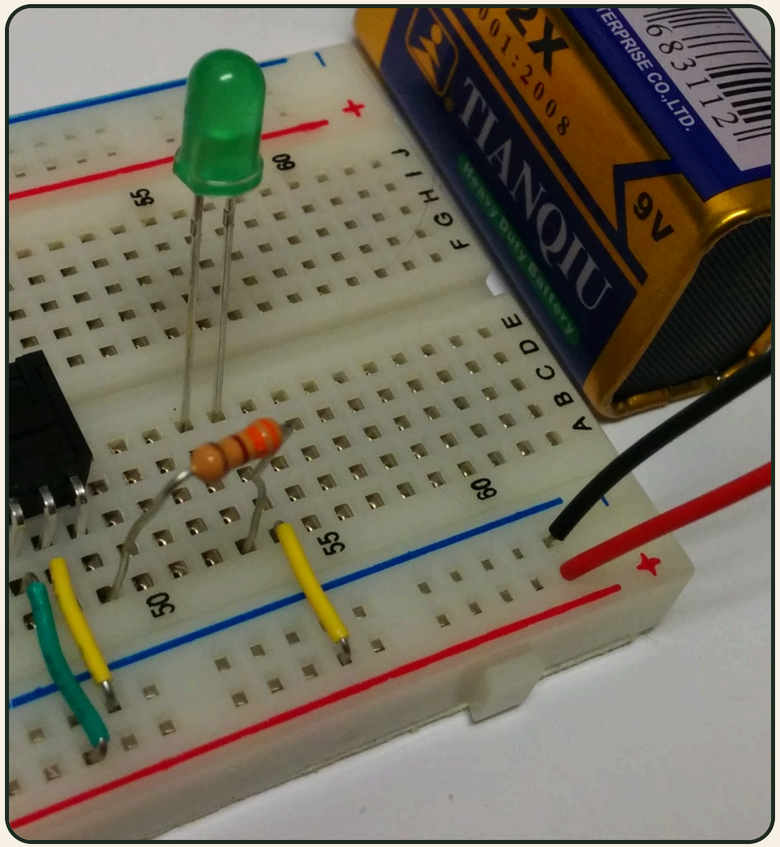


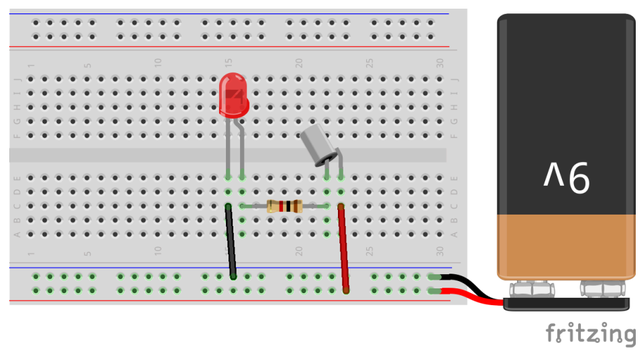
SWITCHES

Tilt Ball Switch

Emergency on/off if project tips over.



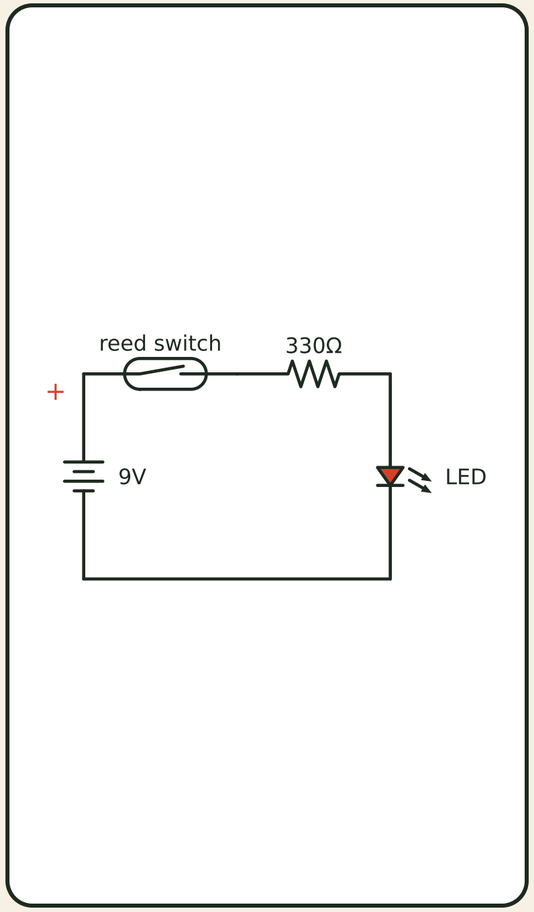


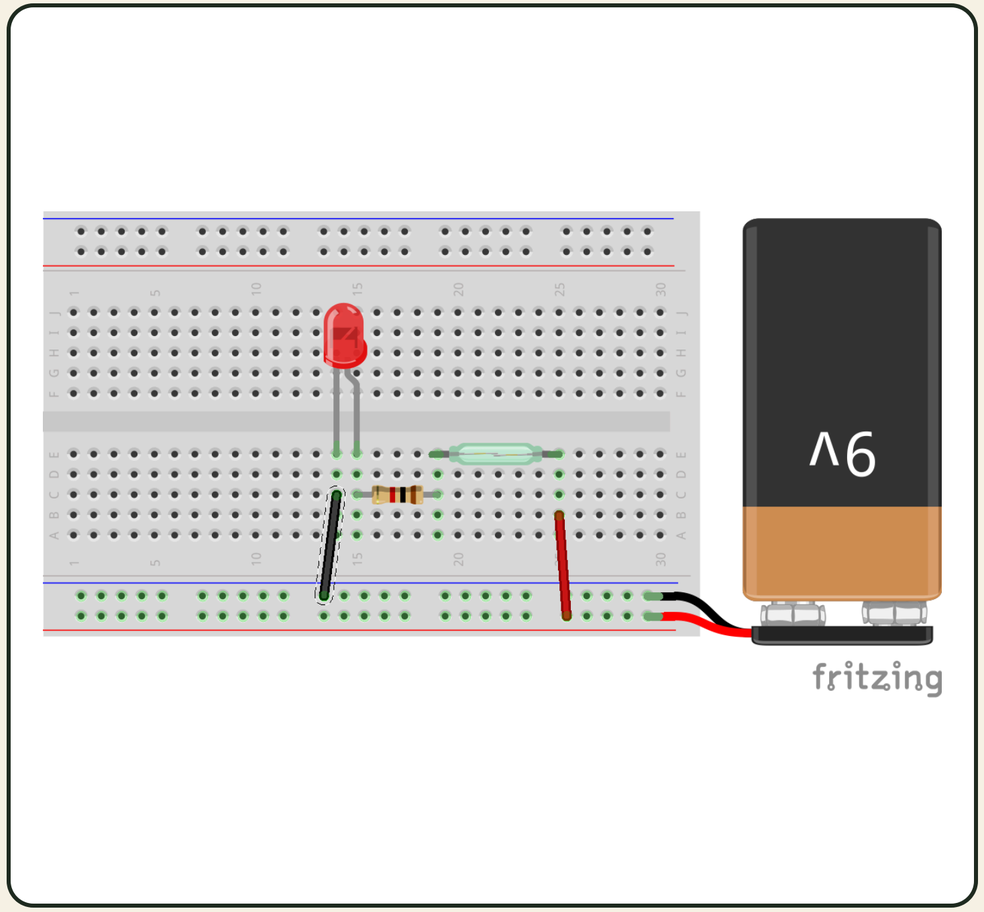


Reed Switch Example

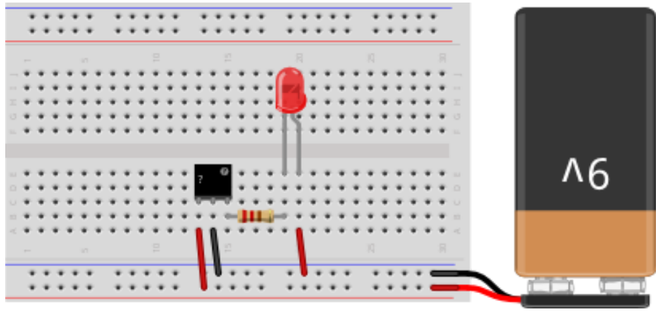
A reed switch is closed when a magnet is in close proximity.

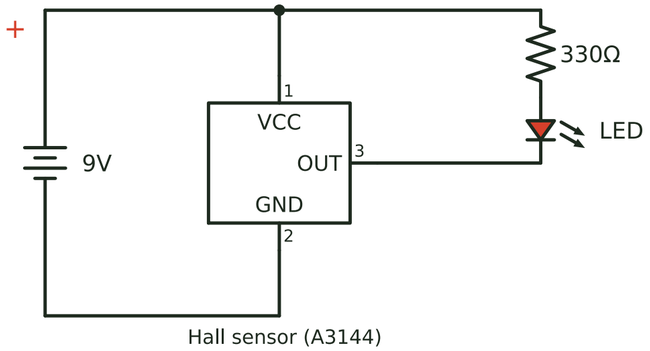
It is symmetric so it can be placed either direction.

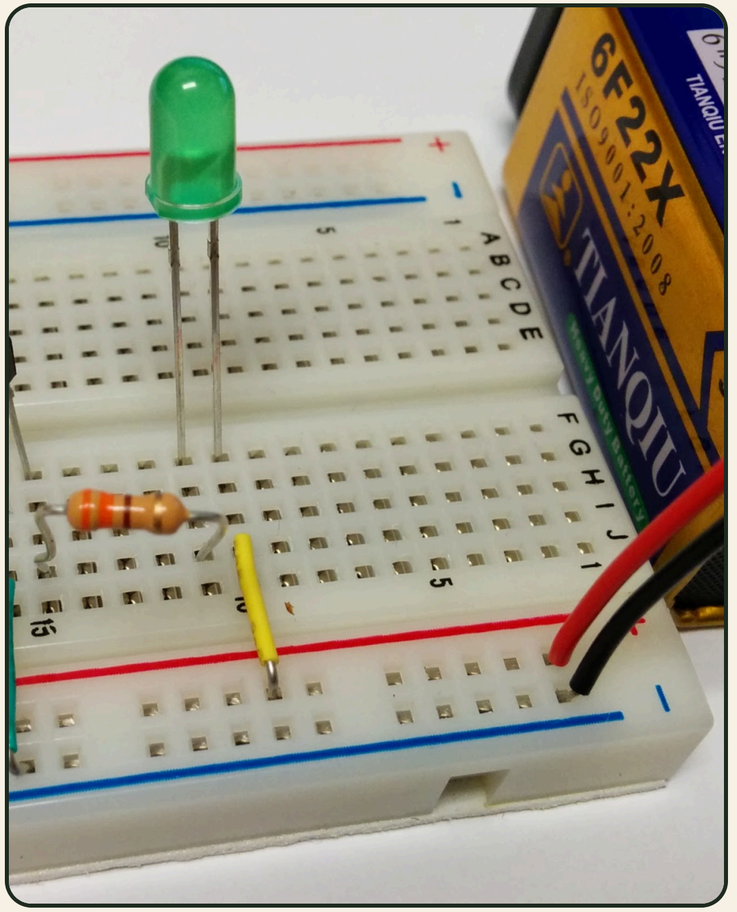




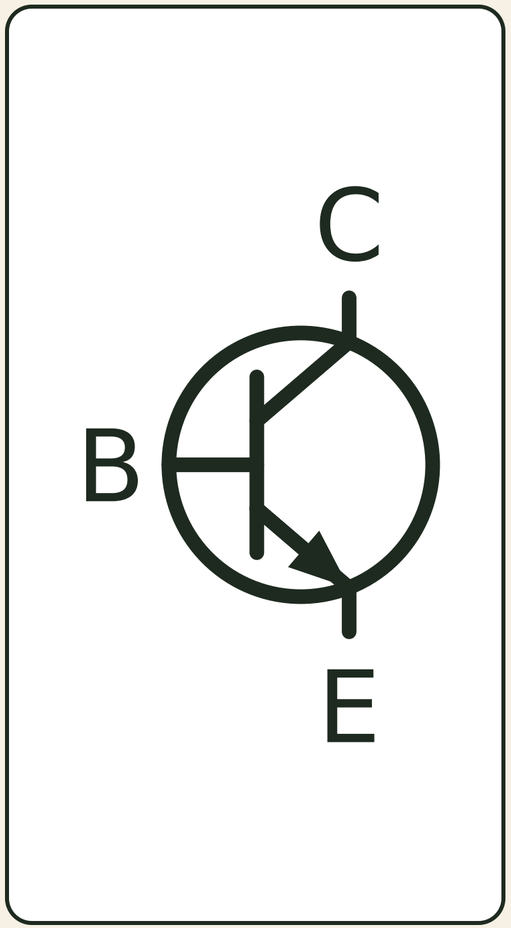
Hall Effect Sensor

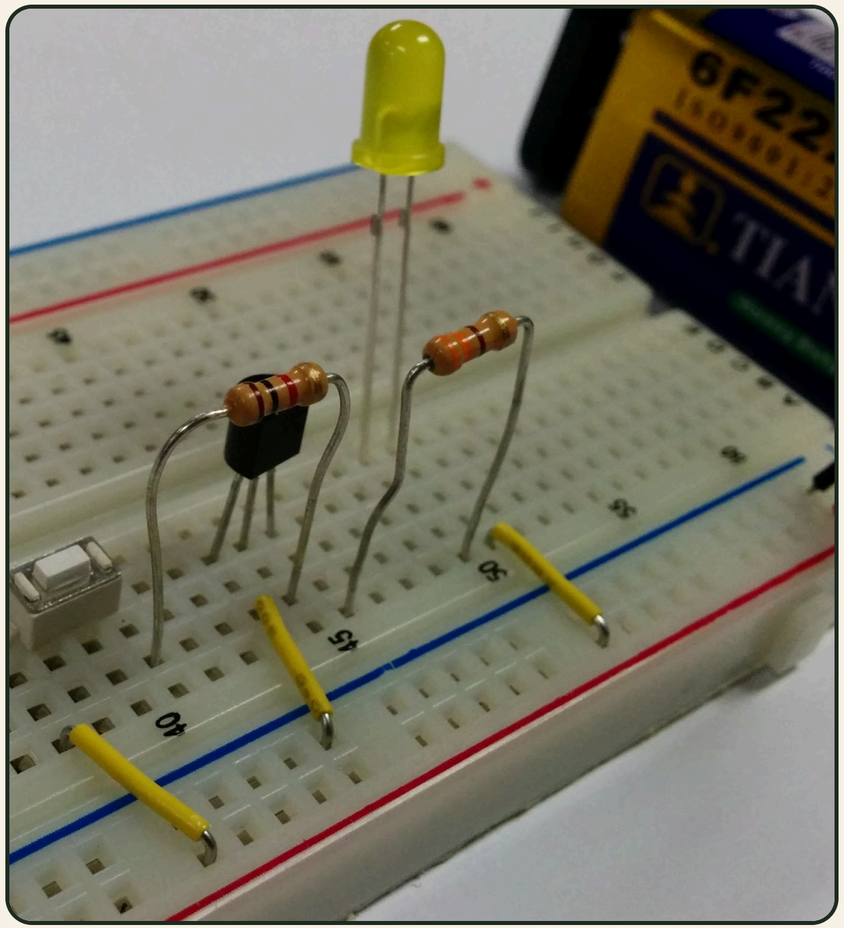


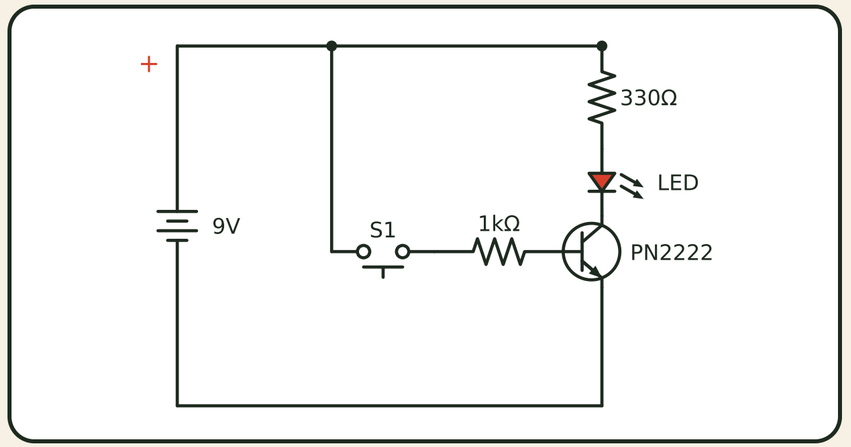


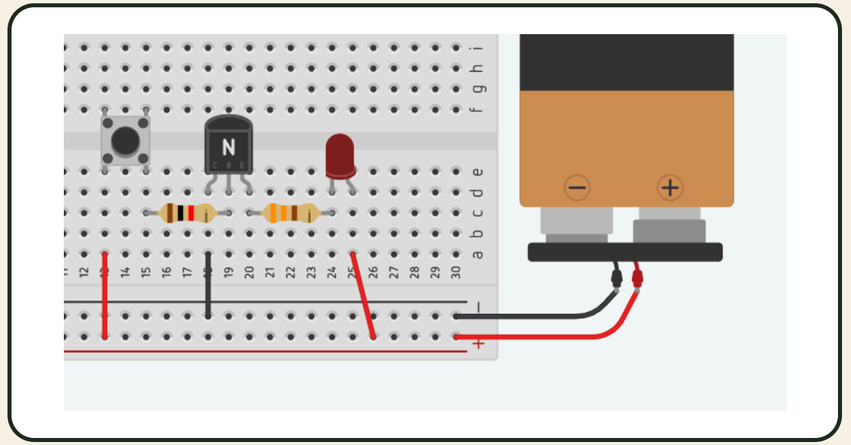


NPN Transistor

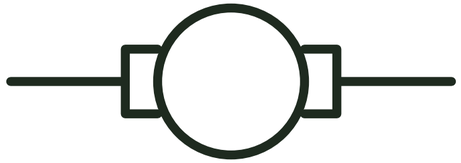


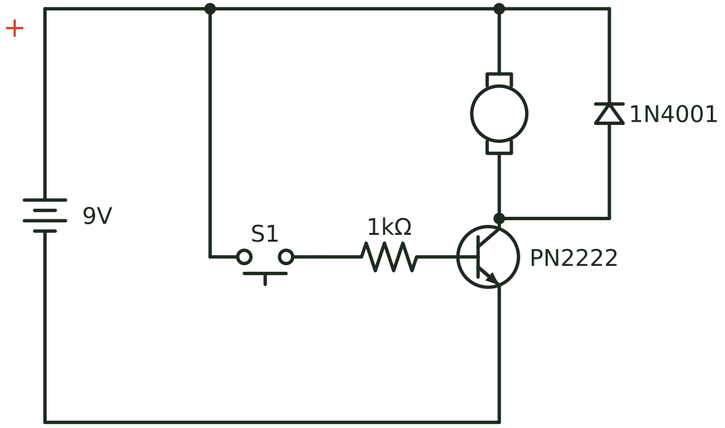


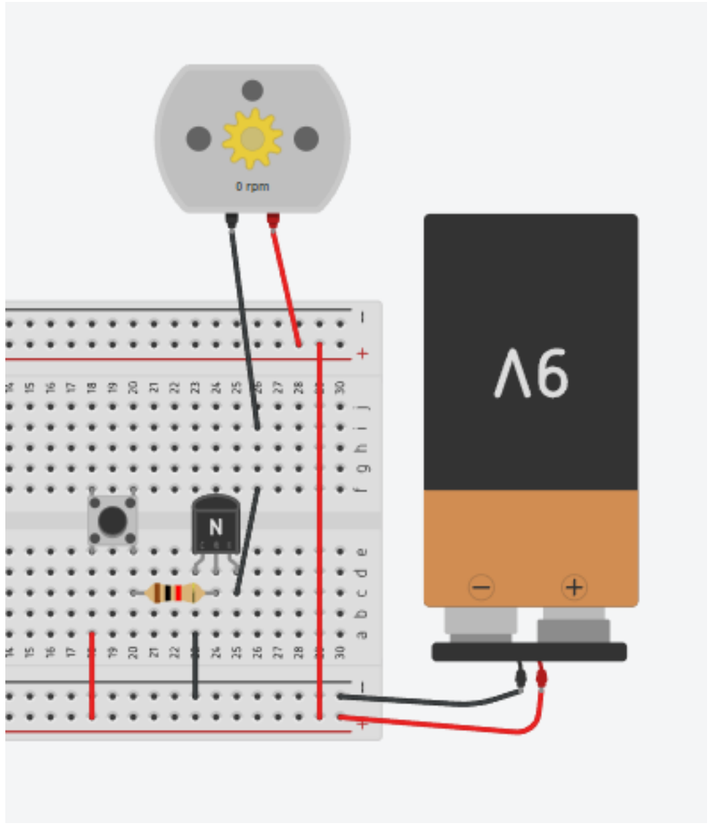




Motors







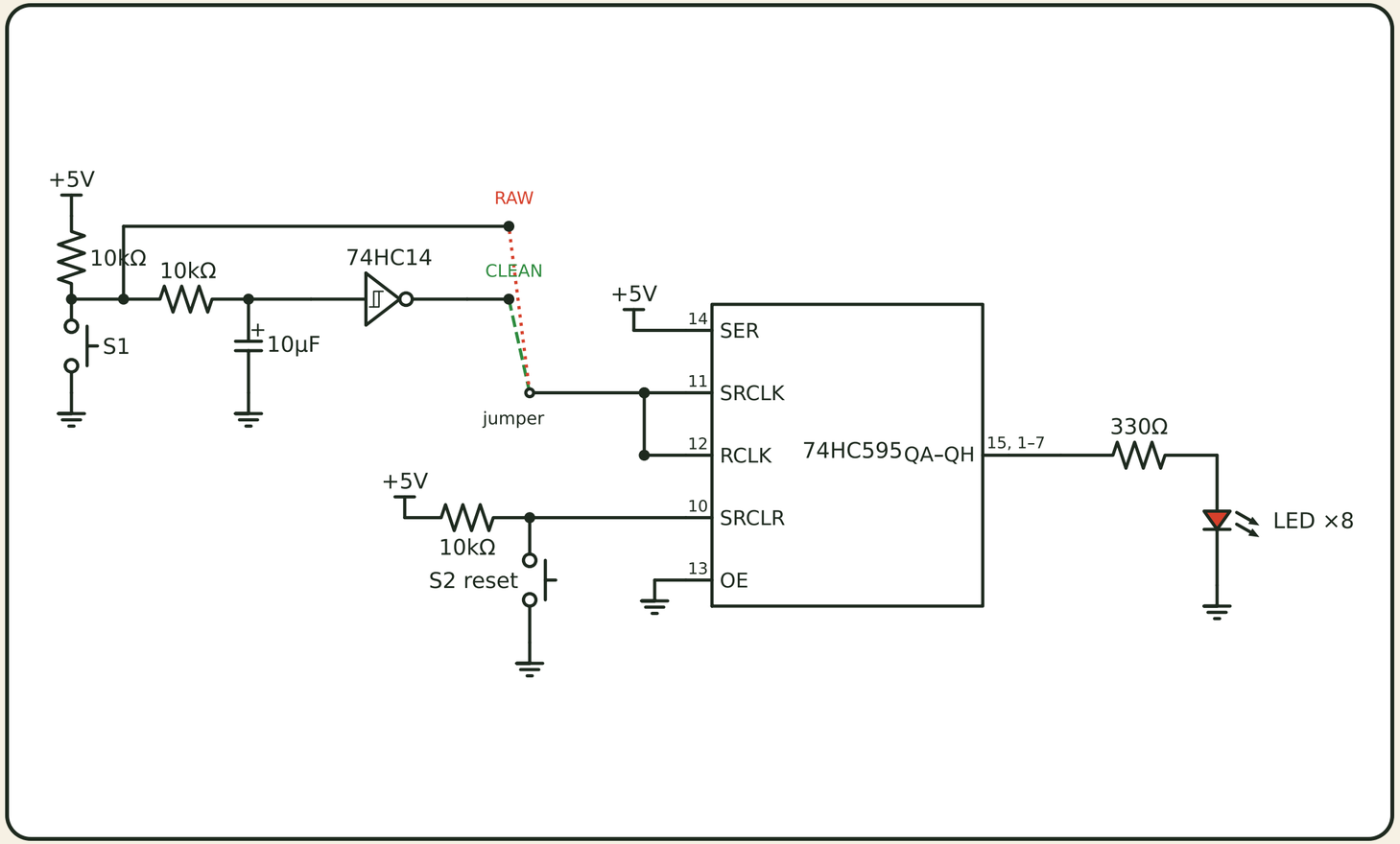
Count the Bounces

Bonus

A button's metal contacts bounce. Each bounce shifts one more lit LED into the 74HC595.

ACTIVITY

1. Jumper on **RAW**. Press S2 to clear, then tap S1 once.
2. Count the LEDs. More than one means the button bounced.
3. Move the jumper to **CLEAN** and repeat. One tap, one LED.



5V only. 9V destroys these chips. Use a 5V breadboard power module or USB. Wire each chip's power: 74HC595 pin 16 → +5V, pin 8 → GND; 74HC14 pin 14 → +5V, pin 7 → GND. Tie unused 74HC14 inputs to GND.

REFERENCE

Schematic Symbols



Resistor

Limits current. Color bands give its value.



Potentiometer

A resistor with a knob. The arrow is the slider.



Capacitor

Stores a little charge. Fits either way round.



Electrolytic cap

Stores more charge. + leg goes to the + side.



Diode

One-way valve. The bar is the – side.



LED

A diode that glows. The long leg is +.



Battery

Power source. The long plate is +.



Ground

The 0V return path, the – side.



Supply

A connection to the + supply voltage.



Junction

A dot: these wires are connected.



Crossing

No dot: the wires pass without touching.



Switch

Stays on or off until you move it.



Push button

On only while you hold it down.



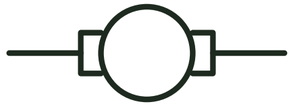
Reed switch

Closes when a magnet comes near.



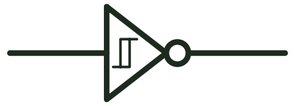
NPN transistor

A small base current switches a bigger one.



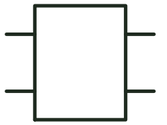
Motor

Turns electricity into spinning.



Schmitt inverter

A NOT gate that ignores wobbly inputs.



Chip (IC)

Pins are numbered on its datasheet.